

Sandwich Ontology

1. Overview

The ontology models the structure of sandwiches that captures the whole (Sandwich), its parts (BreadSlice, LayerOfFilling, Filling), and six variants (SingleDeckerSandwich, DoubleDeckerSandwich, OpenSandwich, ToastedSandwich, Panini, ClubSandwich).

I also created five object properties describe how these elements are related: hasBreadSlice that links a sandwich to each slice of bread; hasLayer links a sandwich to each vertical layer of filling; hasFillingIngredient links a layer to any filling substance (hasMeatFilling and hasVegetableFilling).

2. Justification

For classes, Sandwich is placed directly under owl:Thing; everything that is a sandwich inherits from it.

Then, SingleDeckerSandwich, DoubleDeckerSandwich, OpenSandwich, ToastedSandwich, Panini, ClubSandwich are declared subclasses of Sandwich because each one is a sandwich with additional constraints.

BreadSlice, LayerOfFilling, and Filling remain siblings of Sandwich because they are parts of a sandwich, not specialised sandwiches themselves.

For properties, hasBreadSlice and hasLayer are general relations from a sandwich to its components, so they have no parent other than owl:topObjectProperty. hasFillingIngredient is a relation from a layer to any filling. hasMeatFilling and hasVegetableFilling are declared sub-properties of hasFillingIngredient.

3. Axiom

For SingleDeckerSandwich the ontology contains hasBreadSlice exactly 2 BreadSlice and hasLayer exactly 1 LayerOfFilling. This mean every SingleDeckerSandwich must have exactly two bread slices and exactly one deck of filling.

For DoubleDeckerSandwich the ontology contains hasBreadSlice exactly 3 BreadSlice and hasLayer exactly 2 LayerOfFilling. This mean every DoubleDeckerSandwich has exactly three slices and two decks of filling.

4. Advantages and disadvantages

The advantages of this ontology focuses on the structural and compositional aspects of a sandwich, like how many slices of bread it has, how many filling layers, and the distinction between meat and vegetable ingredients. This will help it to keep a clear and focused scope.

The disadvantage of this structure is that there is little further subdivision of the variants. All variants are below Sandwich, it can be harder to capture more detailed groupings or shared behaviour among subsets of sandwiches.